# Optimising Blockchain Voting with Zero-Knowledge Rollup Layer-2

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**Abstract:** In the modern world, traditional electronic voting methods frequently lack transparency and verifiability, which compromises the integrity of elections and diminishes public trust. Therefore, blockchain technology offers a decentralized and safe way to vote online. However, current implementations have problems with privacy protection, scalability, and higher transaction costs. This dissertation presents an enhanced blockchain voting framework that integrates Zero-Knowledge Proofs (ZKPs) within a zk-Rollup Layer-2 architecture. The suggested solution uses Zero-Knowledge Proofs (ZKPs) to keep voters' identities secret and make sure that votes can be verified as accurate. zk-Rollups, on the other hand, do off-chain aggregation and send short proofs to the main chain. This combination will make things more efficient, scalable, and private, and it will also help the development of safe, open, and trustworthy electronic voting systems that encourage people to get involved in democracy.

*Keywords:* Blockchain Voting; Zero-Knowledge Proofs; zk-Rollup;

*Disclaimer:* I declare that this dissertation is entirely my own work and has not been submitted, in whole or in part, for any other academic award. All sources of information have been properly acknowledged through citation and referencing in accordance with academic standards.

## 1. INTRODUCTION

### *1.1 Background*

The integrity of political processes depends on having voting systems that are transparent, secure, and reliable. Traditional paper-based elections are generally trusted, but they can be slow, inefficient, and prone to human error [46]. As societies continue to adopt digital infrastructures, *electronic voting (e-voting)* has emerged as a faster and more accessible alternative [12]. However, centralized systems are still vulnerable to malicious attacks, such as Distributed Denial-of-Service (DDoS), and introduce single points of failure that undermine both system availability and trust [57]. If a central server is compromised or goes down on election day, ballots could be exposed, altered, or even lost without detection, highlighting how fragile centralized architectures can be in contexts where trust is critical[52]. These limitations have led researchers to explore *blockchain technology* as a decentralized, tamper-resistant foundation for secure digital voting. By using distributed consensus, blockchain allows independent verification and transparent auditing, which in turn strengthens institutional trust and accountability—key attributes of electoral integrity in modern democracies [49].

Despite its great potential, the practical adoption of blockchain for large-scale elections remains limited by critical technical barriers that must be addressed to ensure viability in national-scale deployments. In particular, the capacity to manage millions of votes effectively and complex privacy demands presents significant problems [36]. In addition, significant computing requirements and additional technical constraints impede the nationwide implementation of blockchain voting [28, 36]. Resolving these difficulties is essential to guarantee that a blockchain-based voting system is feasible and reliable for national elections.

### *1.2 Problem Statement and Motivation*

Current blockchain-based voting systems, especially those operating directly on Layer-1 (L1) networks still face three major challenges: limited scalability, the need to balance privacy with transparency, and high operational costs.

(1) The Privacy–Transparency Dilemma
The secret ballot, requiring complete voter anonymity, is a core principle of democratic elections. Public blockchains, however, achieve transparency and verifiability by making all transaction data publicly visible. This creates an inherent conflict with the need to keep ballots confidential [54]. Existing cryptographic methods aimed at protecting voter anonymity while maintaining transparency often come with heavy computational costs, which further worsen scalability issues [36]. Moreover, the central challenge is to verify that each voter is eligible and votes only once, without allowing the final vote to be traced back to the voter's identity [53].

(2) Scalability Constraints
The throughput limitations of foundational L1 blockchains render them unsuitable for high-turnout national

elections, which require handling millions of transactions (votes) within very short time windows [36]. Systems using Proof of Work (PoW) consensus suffer from low transaction throughput and high latency due to computational bottlenecks. While more energy-efficient models like Proof of Stake (PoS) and Delegated Proof of Stake (DPoS) offer improved efficiency, they still struggle with the volume required for large elections and introduce risks such as centralization and potential collusion [36]. For blockchain technology to support democratic processes at scale, a solution must provide high throughput far exceeding L1 capacity.

(3) Cost Prohibitions
Submitting each ballot directly on a Layer 1 blockchain such as Ethereum results in notably high transaction fees, especially during times of elevated network activity [52]. For instance, a standard Ethereum L1 transaction often exceeds $1.10 in fees, whereas typical costs on major Layer-2 (L2) networks are usually below $0.10 [38].

Moreover, direct L1 implementation presents distinct technical cost obstacles for elections. Executing intricate elliptic curve operations—essential for secure tallying algorithms such as Homomorphic Encryption—within an L1 smart contract is so resource-intensive that a single contract can count a maximum of approximately 40 votes before exhausting the entire block gas limit [48]. Additionally, storing the requisite list of eligible voters directly on the L1 smart contract is prohibitively expensive. The `SSTORE` opcode incurs a cost of approximately 20,000 gas to store 32 bytes of non-zero data, meaning that recording even 500 voters' addresses can consume around 10 million gas—a substantial fraction of the current Ethereum block gas limit [48]. Consequently, these high storage and computation demands make large-scale voter registration on L1 impractical, effectively preventing blockchain voting systems from supporting nationwide participation.

Motivation: The primary challenge in implementing national-scale blockchain voting lies in simultaneously addressing three core constraints: scalability, privacy, and cost. Achieving this balance requires a carefully designed architecture: ZK-Rollup Layer-2 systems reduce cost and scalability issues by handling most transactions off-chain [33], while still relying on Ethereum L1 for security, and Zero-Knowledge Proofs address the privacy–transparency trade-off by proving voter eligibility and ballot validity without exposing any private data [40]. This research is motivated by the need for a trust-minimized, scalable, and cost-efficient voting framework that leverages Ethereum Layer-1 security through a ZK-Rollup Layer-2 design.

### *1.3 Scope*

The research scope is precisely defined by the requirements for a secure electoral system and is structured to address the specific comparisons posed by the research questions. This study specifically examines the Polygon zkEVM ZK-Rollup Layer-2 architecture, which is solely anchored to Ethereum Layer 1 as the definitive trust and data availability layer. The principal parameters delineating the scope are:

- Comparative Benchmark: The project will directly compare the cost, throughput, and latency of the proposed Zero-Knowledge Election Rollup (ZKER) Layer-2 design with a simulated traditional Ethereum Layer-1 system, generating the quantitative data needed to answer RQ3.
- Cryptographic Focus: The study is limited to integrating Zero-Knowledge Proofs for anonymity and Homomorphic Encryption for confidential tallying within the ZK-Rollup environment, thereby addressing the core mechanisms tested in RQ2.
- Functional Context: The application context is restricted to a single-choice, high-volume election, targeting a throughput benchmark of more than 71 transactions per second (TPS) to demonstrate feasibility for mass adoption [25].
- Outcome: The outcome will be an independently replicable Proof-of-Concept (PoC) architecture and empirical performance metrics, ensure the work is verifiable and can be expanded for future practical decentralized governance applications.

### *1.4 Research Questions*

This research will address the following primary research questions:

**RQ1:** How can zero-knowledge proofs combined with a zk-rollup Layer-2 architecture improve the scalability, voter privacy, and verifiability of blockchain-based electronic voting systems?

**Methodology:** Design and implement the ZK-Verified Election Rollup (ZKER) Proof-of-Concept on Polygon zkEVM (L2), explicitly separating high-volume execution (L2) from trust verification (L1) [14]. This requires implementing the L2 sequencer's transaction batching and the aggregator's zero-knowledge proof generation for large volumes of encrypted votes. The final validity proof is then sent to the Ethereum L1 verifier smart contract [39]. This flow of a zk-Rollup layer is showed in Figure 1.

**Expected Outcome:** The implementation is expected to produce empirical evidence demonstrating architectural enhancements, validated through the following these metrics:

- Scalability Metrics: Measure the efficiency of Layer-2 aggregation using *Batch Size* and *Votes-per-Proof*, capturing the number of ballots processed per cryptographic proof.
- Data Efficiency: Quantify the reduction of computational and storage load on Layer 1 using the *Compression Ratio (L2:L1 Data)*, comparing the total transaction volume executed on Layer 2 against the final proof and state root committed to Layer 1.
- Verifiability Integrity: Evaluate the functional correctness and reliability of the security anchor by tracking *Proof Verification Success Rates* on the Layer 1 Verifier Contract.
- Architectural Anonymity: Confirm that no data linking a voter's identity to their ballot is exposed on the public Layer 1 chain, validating the zk-rollup

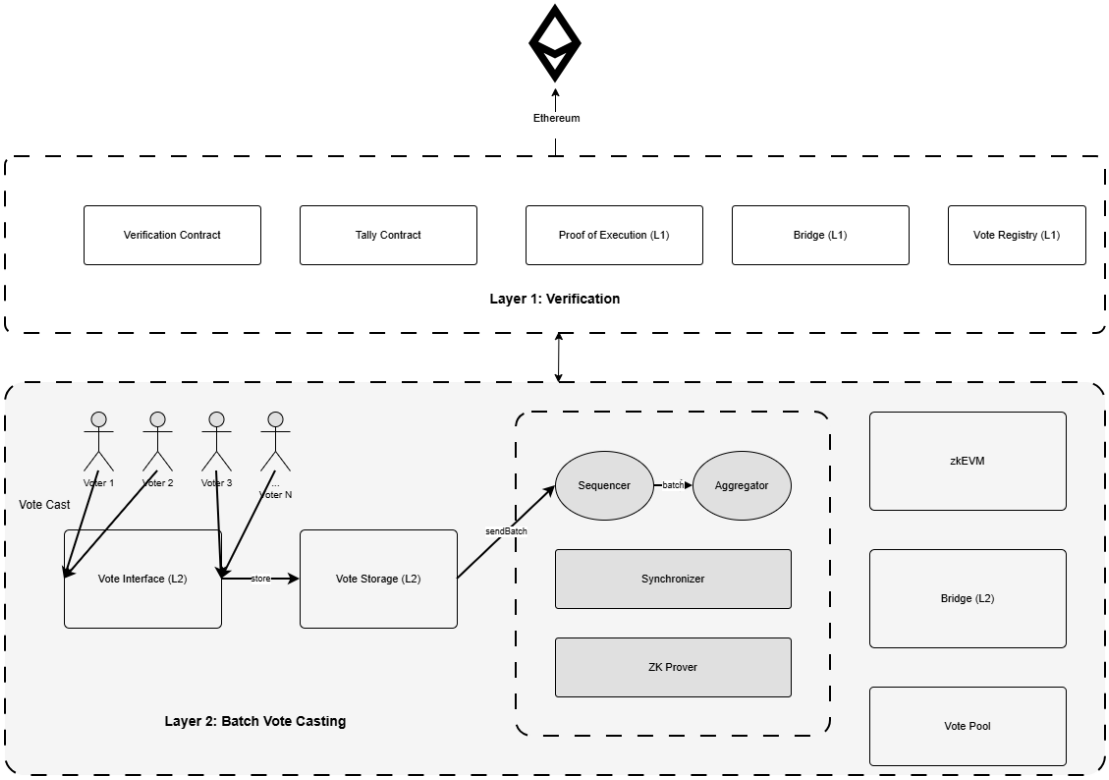



Fig. 1. zk-Rollup-based voting workflow that includes batching, proof generation, and Layer-1 verification.

architecture's privacy guarantees through qualitative evaluation.

**RQ2:** How effectively can zero-knowledge proofs guarantee voter anonymity and ballot secrecy without compromising end-to-end verifiability?

**Methodology:** Develops a technique for privacy-preserving electronic voting utilizing Zero-Knowledge Proofs and zk-Rollups, organized into three cryptographically safe and privacy-preserving phases [3, 42]:

1. Authentication and Uniqueness: Each voter holds a secret $sk_v$ used to generate an identity commitment $\mathsf{com}_v = \mathsf{Com}(sk_v, \mathsf{aux}_v)$, which is stored in the Layer 1 Merkle root. The voter produces a zero-knowledge proof demonstrating that $\mathsf{com}_v$ is a valid leaf of the Merkle root (eligibility) and derives a unique nullifier [21] to prevent double voting, without revealing identity.

$$\mathsf{Nullifier} = H(sk_v \parallel \text{ElectionID})$$

2. Ballot Encryption and Proof Generation: The voter selects a candidate $B$ and encrypts it using a verifiable additively-homomorphic encryption scheme with randomness $R$. A zero-knowledge proof $\mathsf{ZKP}_{\text{ballot}}$ confirms that the ciphertext was correctly computed and that $B \in \mathcal{C}_{\text{valid}}$, without revealing $B$.

$$C = \mathsf{Enc}_{\mathsf{pk}_E}(B; R)$$

3. Homomorphic Tallying and Verifiable Decryption: All valid ciphertexts $\{C_i\}$ are aggregated using the homomorphic property:

$$C_{\text{Total}} = C_1 \oplus C_2 \oplus \cdots \oplus C_n$$

A threshold group of trustees jointly performs decryption via Distributed Key Generation (DKG) [50] and publishes a zero-knowledge proof $\mathsf{ZKP}_{\text{dec}}$ confirming that the final plaintext tally was correctly decrypted without revealing individual votes.

**Expected Outcome:** The implemented protocol will generate securely protected ballots whose integrity can be publicly verified, evaluated using the following metrics:

- Nullifier validation
  - Test data includes private inputs ($sk_v$, $B$) used to generate the Nullifier and ZKP of Merkle membership/uniqueness. [20, 23]
  - Expectation: Demonstrate the L2 Ballot Contract successfully prevents double voting by checking the Nullifier registry.
- Integrity Testing (Malicious Input) [31]
  - Test data includes invalid ciphertexts, reused nullifiers, or ballots with $B \notin \mathcal{C}_{valid}$.
  - Expectation: Malicious votes (double voting, invalid candidates, or malformed proofs) are automatically rejected.
- End-to-End Verifiability
  - Test Procedure: Demonstrate the post-election verifiable tally through Layer 2 homomorphic aggregation ($C_{\text{Total}}$), followed by collective decryption using Distributed Key Generation (DKG) with threshold key holders [32].
  - Expectation: Test the final ZKP verification process submitted to Layer 1. The test confirms that any observer can verify the final tally using only public data (the ZKP proof, $C_{\text{Total}}$, and the DKG output) [19]. The proof must confirm that: (1) the aggregation of encrypted ballots is correctly computed, (2) the underlying ballot proofs are valid, and (3) the DKG threshold decryption was correctly performed [37].

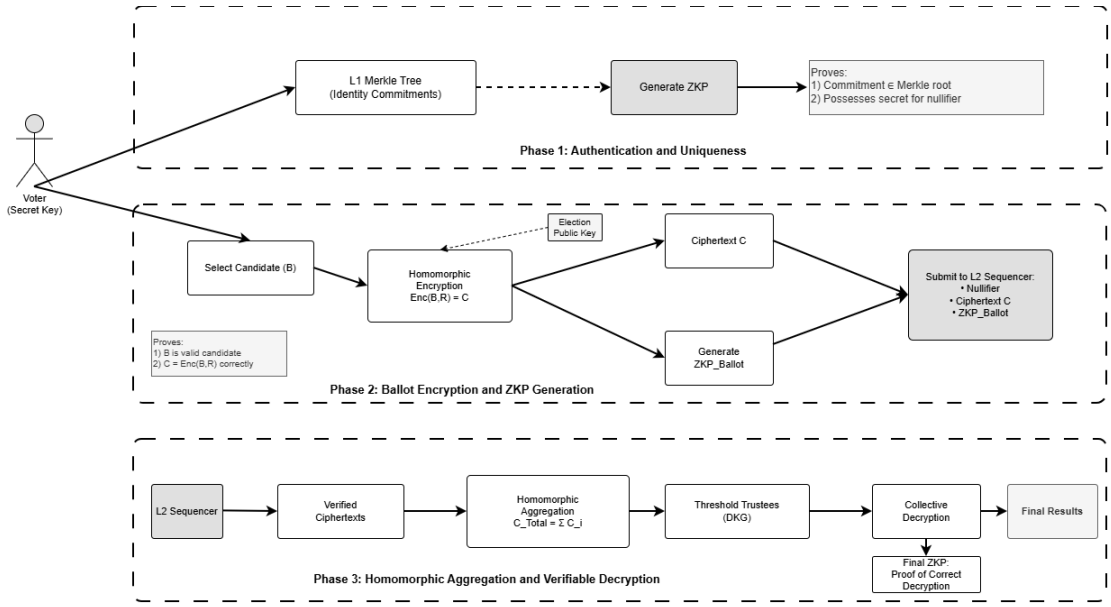



Fig. 2. Privacy-preserving verification using Zero-Knowledge Proofs.

**RQ3:** To what extent does a zk-rollup-based voting system reduce transaction costs and increase throughput compared to a traditional Layer-1 blockchain voting model?

**Methodology:** Execute simultaneous empirical benchmarking of transaction costs and performance across two environments: (A) the proposed ZKER system implemented on Polygon zkEVM (L2) (B) a baseline voting model simulated on Ethereum Layer 1 (L1).

*Experimental Setup:* Use a dedicated load simulation environment to inject identical, high-volume batches of synthetic vote transactions (e.g., 10,000 votes per batch) into both the L1 and L2 environments under varying stress conditions. [25]

*L1 Baseline Implementation:* A simple smart contract records encrypted votes sequentially in a simulated Ethereum L1 environment, providing a reference for gas consumption and throughput limitations.

*L2 Proof-of-Concept (PoC) Implementation:* The ZKER system will handle the same transaction load on Polygon zkEVM, measuring both the total Layer-1 gas cost for proof verification and the overall Layer-2 transaction fees.

**Expected Outcome:** The comparison will show that:

- Throughput: Compare Transactions Per Second (TPS) achieved in both environments.
- Cost Efficiency: Compare the average cost per vote.
- Summary Table: Compile the results into a comparison table showing TPS, latency, and average transaction cost for both setups, addressing the main evaluation goal of RQ3.

**RQ4:** Which ZKP scheme (zk-SNARKs, zk-STARKs, Bulletproofs) provides the optimal balance of performance and security for a Layer-2 rollup voting system?

**Methodology:** Perform a comprehensive and literature-based analysis of the three principal Zero-Knowledge Proof scheme frameworks zk-SNARKs, zk-STARKs, and Bulletproofs—focusing on their respective trade-offs within the framework of ZK-Rollup electoral systems.

**Expected Outcome:** Following the analysis of the key performance indicators for all three schemes, a comparison matrix will be constructed [11, 27]:

- Proving Time: Time required for the Prover (voter's client or L2 aggregator) to generate the ZKP.
- Verification Time: Time required for the L1 Verifier Contract to check the proof (critical for L1 gas cost).
- Proof Size: The cryptographic size of the proof, directly influencing L1 data commitment cost (gas).

**RQ5:** How does a zero-knowledge voting system behave across different EVM-compatible zk-rollup environments (e.g., Polygon zkEVM vs zkSync Era), and what architectural differences most significantly influence performance, cost, and security?

**Methodology:** Conduct a comparative architectural analysis of Polygon zkEVM and zkSync Era to discern characteristics influencing e-voting performance, cost, and security. The ZKER proof-of-concept on Polygon zkEVM serves as the primary reference, whereas zkSync Era is assessed using literature and public benchmarks.

**Expected Outcome:** Compare the performance impact of these architectural choices on the target E-Voting key performance indicators [4, 56]:

- Transaction Cost: Compare the Average L2 Transaction Fee for simple transactions (proxy for vote submission) and the subsequent L1 Proof Verification Cost.
- Scalability: Compare published Throughput (Transactions Per Second) figures and Batch Sizes.

## 2. PRELIMINARY LITERATURE REVIEW

This section will examine the supporting research literature pertinent to the proposed study, focusing on main key areas: blockchain voting systems, zero-knowledge proofs, and zk-rollup Layer-2 architectures.

### 2.1 The Genesis of Blockchain Voting

The foundation of this research area is Satoshi Nakamoto's 2008 whitepaper, *"Bitcoin: A Peer-to-Peer Electronic Cash System"* [44]. Although presented in a financial context, Nakamoto's real breakthrough was offering a practical solution to the Byzantine Generals Problem [45] through Proof-of-Work (PoW). This "trustless" feature is new in the field of e-voting literature; it seems to replace the error-prone human election administrator with statistical certainty. Thus, the literature highlights blockchain's immutability as its key protection against long-standing risks such as ballot box stuffing or tampering with election results after the fact [28].

### 2.2 The Rise of Smart Contracts and the EVM

The shift from simple distributed calculators to fully distributed computers was largely propelled by Vitalik Buterin's 2013 Ethereum whitepaper, *A Next-Generation Smart Contract and Decentralized Application Platform* [9]. In this work, Buterin made blockchain technology more useful by adding the Ethereum Virtual Machine (EVM), which is almost Turing-complete and can run any program, including smart contracts.

The concept of the smart contract exists prior to blockchain technology. The literature identifies Nick Szabo's seminal 1996 article, *Smart Contracts: Building Blocks for Digital Markets* [51], as the origin of the term. Szabo characterized a smart contract as "a collection of commitments, articulated in digital format, encompassing protocols through which the parties fulfill these commitments." This theoretical groundwork supports the dissertation's premise that *election law* can be expressed through smart contract logic, making electoral rules to become self-enforcing, transparent, and resistant to tampering.

In the modern world, some standard technical sources, such as Mastering Ethereum by Andreas Antonopoulos and Gavin Wood [6], offer detailed insights into how smart contracts are deployed and executed on the Ethereum platform. These works outline key concepts such as gas fees, the EVM's structure, and on-chain data storage—factors that directly affect the cost and practical feasibility of implementing voting systems on a blockchain. In particular, the SSTORE opcode, which is responsible for writing data to persistent storage on Ethereum Layer 1, incurs extremely high gas costs [48]. This makes storing large datasets, such as voter registration records, economically impractical, thereby motivating the shift toward off-chain or Layer 2 architectural approaches.

### 2.3 Electronic Voting Protocols

*Historical Context* The seminal book *Real-World Electronic Voting: Design, Analysis and Deployment* [30] by Feng Hao and Peter Y. A. Ryan provides a thorough historical overview by highlighting the shortcomings of early electronic voting systems, including punch-card machines implicated in the "hanging chad" [15] controversy during the 2000 U.S. Presidential Election [17]. This failure led to the widespread adoption of Direct Recording Electronic (DRE) systems [16].

However, Direct Recording Electronic (DRE) systems introduced the "black box" problem. According to Halderman (2016), DRE systems are essentially computers running proprietary software that voters cannot inspect. If the software contains a flaw or is malicious, it can alter votes without detection [29]. This vulnerability led to the academic pursuit of *Software Independence* and *Voter Verified Paper Audit Trails (VVPAT)*, concepts advocated by Rivest and Wack and further elaborated in the National Academies' *Securing the Vote* report.

*End-to-End Verifiability* To address the "black box" problem without relying solely on paper-based mechanisms, cryptographers developed the concept of *End-to-End Verifiability*. The literature defines E2E verifiability through two core properties:

(1) Individual Verifiability: A voter can cryptographically verify that their ballot was cast as intended and correctly recorded on the public bulletin board.
(2) Universal Verifiability: Any observer can cryptographically verify that the final tally is the correct aggregation of all the recorded ballots.

Table 1. Comparison of Key Cryptographic Voting Constructions

| **Article / Concept** | **Key Contributions, Pros and Cons** |
|---|---|
| **Chaum (1981) – Mixnets** [13] | **Pros:** Introduced anonymity through shuffled and re-encrypted messages, breaking the link between voter and vote. Foundation for privacy-preserving e-voting. **Cons:** Does not address coercion or vote-buying; allows a voter to prove how they voted if a receipt is available. |
| **Benaloh & Tuinstra (1994) – Receipt-Freeness** [8] | **Pros:** Prevents voters from proving how they voted, blocking vote-buying and voluntary coercion. Formalises the idea that proofs of vote should be impossible. **Cons:** Does not protect voters during supervised coercion; coercer can still force action. |
| **Juels, Catalano & Jakobsson (2005) – Coercion-Resistance / Fake Credentials** [35] | **Pros:** Strongest protection; coerced voter can simulate compliance using fake credentials, while real vote stays secret and counted. Effective in supervised voting scenarios. **Cons:** More complicated, more protocol overhead, and harder to put into place. Needs stronger trust assumptions and better cryptographic infrastructure. |

*Blockchain-based Voting System* Modern literature increasingly regards blockchain as a viable solution for comprehensive voting systems [28, 54]. A broad review of blockchain-based e-voting shows that blockchain provides immutability and transparency missing in traditional centralized systems. But it also shows a big problem: making votes public makes it easier to check them, but it makes it much harder to protect the privacy of voters. Addressing this trade-off requires advanced cryptographic tools, such as zero-knowledge proofs—techniques that my dissertation relies on to preserve privacy while still ensuring transparency and trust in a blockchain-based voting system.

### 2.4 PoW vs. PoS

The fundamental security framework of blockchain is under intense scholarly examination. The transition of Ethereum from Proof-of-Work (PoW) to Proof-of-Stake (PoS) during *The Merge* (2022) [22] dramatically transformed the environmental and security dynamics of blockchain applications.

Recent surveys and comparative analyses, such as those by Leporati and Rovida (2024), investigate the stability and fairness of Proof-of-Stake (PoS) consensus [43]. Although Proof-of-Work (PoW), as used in Bitcoin, offers strong *thermodynamic security* because of its high energy requirements, it has been widely criticized for its environmental impact, making it difficult to justify for government-run voting systems. In contrast, PoS reduces energy consumption by more than 99%, removing a major barrier to adoption. Even so, the literature warns that PoS introduces its own risks, particularly the potential for increased centralization.

As the "rich-get-richer" phenomenon, where those with large stakes gain more power, raises concerns about censorship. In a voting situation, a state actor that has most of the stake might, at least in theory, stop transactions from voters who don't agree with them. This kind of approach keeps the security guarantees of the foundational layer while lowering its operating costs [2]. In my dissertation, ZK-Rollups use cryptography instead of economic incentives to make sure things are correct. Every state update includes a zk-SNARK or zk-STARK proof that verifies all underlying transactions, and Ethereum's verifier contract approves or rejects updates based solely on that proof. Since validity is determined mathematically rather than by validator agreement, even a powerful PoS stakeholder cannot alter the state or forge votes.

### 2.5 Zero-Knowledge Proofs

*Foundation of ZK* Zero-Knowledge Proofs were first introduced in 1985 by Goldwasser, Micali, and Rackoff [26]. They demonstrated that an individual (the *Prover*) can substantiate their knowledge of particular information—such as a legitimate vote—without disclosing any confidential details. The essential concept is the *Simulation Paradigm*: if the verifier can reconstruct the proof independently of the Prover, then no genuine knowledge has been disclosed. Subsequently, in 1986, Fiat and Shamir proposed a technique to convert interactive proofs into *non-interactive proofs* (NIZKs) [24]. This is essential for blockchain and digital voting, as a voter may produce a proof on their device and submit it directly to a smart contract without requiring live interaction.

*zk-SNARKs vs. zk-STARKs* Current literature emphasizes the optimization of these proofs to comply with blockchain limitations, including gas costs and verification duration. The proposed research necessitates a comparative analysis of proof systems (RQ4).

Application to Voting: The paper *Zerocash: Decentralized Anonymous Payments from Bitcoin* (Ben-Sasson *et al.*, 2014) [7] applied zk-SNARKs to protect financial privacy and introduced the *Commitment–Nullifier* paradigm. In this model, the *commitment* represents a voter's eligibility, and the *nullifier* prevents double voting without exposing their identity.

Table 2. Comparison of zk-SNARKs and zk-STARKs

| **Feature** | **zk-SNARKs** | **zk-STARKs** |
|---|---|---|
| References | *The MoonMath Manual to zk-SNARKs* [41] and *Proofs, Arguments, and Zero-Knowledge* [55] by *Justin Thaler* | Pioneered by Eli Ben–Sasson and colleagues (founders of StarkWare) [18] |
| Full Name | Succinct Non-Interactive Arguments of Knowledge | Scalable Transparent Arguments of Knowledge |
| Trusted Setup | Requires a trusted setup (CRS), risk of "toxic waste" | No trusted setup needed (Transparency) |
| Cryptography | Based on elliptic-curve cryptography and pairings | Based on hash functions (post-quantum secure) |
| Proof Size | Very small (typically $<$ 1 kB) | Larger (tens to hundreds of kB) |
| Verification Cost | Fast and efficient | Higher gas cost on Ethereum L1 |
| Scalability | Less scalable for large computations | Highly scalable for large data sets |
| Maturity / Usage | Widely adopted (Groth16, PLONK, zkSync, Polygon) | Emerging adoption (StarkNet, zkSync Era hybrid) |

*Homomorphic Encryption and SAVER* While ZKPs can prove a voter is eligible, tallying encrypted votes calls for homomorphic encryption. With standard encryption, you have to decrypt data before doing any calculations. Additive homomorphic schemes like Paillier or ElGamal avoid this by letting encrypted votes be combined directly, producing an encrypted total that can later be decrypted to reveal the final count:

$$E(A) + E(B) = E(A + B).$$

This allows the blockchain to tally votes without ever accessing the underlying plaintext [34].

However, a malicious voter could try to encrypt an invalid value, like giving one candidate 100 votes. The system needs a way to check that each encrypted vote is real (usually just 0 or 1) without actually decrypting it. This is where SAVER comes in. Lee *et al.* (2019) came up with SAVER, an encryption scheme that works well inside SNARK circuits. This lets you check encrypted votes without letting anyone else see them. It enables *Verifiable Encryption*, ensuring that the ciphertext submitted to the blockchain is well-formed and compliant with election rules [42].

### 2.6 The Layer-2 Architecture

The literature shows that while Layer-1 blockchains offer strong security guarantees, they fall far short of the performance needed for nationwide elections. This limitation is widely discussed within the context of the *Scalability Trilemma*.

*The Scalability Trilemma* In recent systematic reviews [5], the *Scalability Trilemma* asserts that a blockchain architecture can only achieve two of its three core properties at the same time: **Decentralization**, **Security**, and **Scalability**.

- **Decentralization**: Power and verification are spread across many independent nodes rather than concentrated in a single authority.
- **Security**: The level of economic or computational effort an attacker would need to disrupt, censor, or tamper with the ledger.
- **Scalability**: The system's ability to handle a large number of transactions per second in an efficient way.

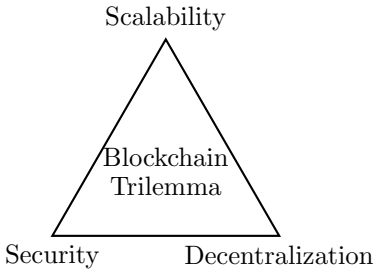



Fig. 3. Blockchain Scalability Trilemma

Ethereum Layer-1 (L1) is architected to prioritize **Decentralization** and **Security**, resulting in a relatively low throughput of approximately 15–30 TPS. For a national election with millions of voters, such throughput is insufficient. For example, processing 10 million ballots at 15 TPS would take:

$$\frac{10{,}000{,}000}{15 \text{ TPS}} = 666{,}666.67 \text{ seconds} \approx 7.7 \text{ days},$$

assuming the network is used only for voting and isn't experiencing congestion. As noted earlier, Ethereum Layer 1 transactions often cost more than $1, while Layer-2 fees are typically just a few cents (L2Fees).

*Layer-2: Rollups* A Rollup is a Layer-2 scaling solution that processes transactions off-chain, consolidates them into a batch, then submits the data or a proof to the Layer-1 blockchain. This architecture enables the L2 to inherit the security of L1 while attaining significantly greater throughput. Vitalik Buterin's blog article "An Incomplete Guide to Rollups" [10] is a foundational document cited extensively in the technical literature. It differentiates between the two principal varieties of rollups:

Table 3. Comparison of Optimistic Rollups and ZK-Rollups

| **Feature** | **Optimistic Rollups** | **ZK-Rollups** |
|---|---|---|
| Examples | Arbitrum, Optimism | zkSync Era, Polygon zkEVM, StarkNet |
| Security Model | Assumes correctness; uses fraud proofs to challenge invalid state transitions | Uses validity proofs (ZKPs) to mathematically prove correctness before finalisation |
| Finality Time | Long finality (up to 7 days) due to dispute window | Fast finality (minutes to hours) once ZKP is verified on L1 |
| Privacy Support | No inherent privacy; requires additional cryptographic layers | Native support for zero-knowledge proofs, enabling voter privacy and ballot secrecy |
| Cost Efficiency | Lower proof generation cost but higher verification gas cost | Higher proof generation cost but lower long-term verification cost and better scalability |
| Suitability for Voting | Limited suitability due to finality delays and weak privacy | Highly suitable: supports cryptographic finality and privacy-first ballot verification |
| Overall Assessment | Suitable for general-purpose scaling but not ideal for election environments | Preferred for E-Voting: aligns with security, privacy, and instant finality requirements. Hence, the literature strongly favors ZK-Rollups for voting applications. |

### *2.7 Polygon zkEVM*

This dissertation focuses on Polygon zkEVM, a Type-2 zkEVM that aims for full EVM equivalence. This makes it easy to use existing Ethereum tools—such as Solidity, Hardhat, and MetaMask—without needing to modify any code.

Recent analysis by Peng et al. (2025) [47] highlights both the strengths and potential risks of this design. Polygon zkEVM uses a *Proof of Efficiency (PoE)* consensus model, where decentralized aggregators compete to produce validity proofs. However, the authors warn that its use of *free inputs* in the constraint system could introduce soundness risks if not formally verified.

The Polygon zkEVM Technical Specification further emphasises that the system defines its STARK constraints using the Polynomial Identity Language (PIL), introducing an additional layer of complexity that the dissertation must carefully address [1].

## 3. TIMELINE

Below is the projected timeline for conducting this research, with each phase aligned to the development and evaluation of the Zero-Knowledge Election Rollup (ZKER) framework. The *Preparation & Literature Review* phase builds the theoretical and technological foundations by analyzing existing blockchain-based voting systems, privacy-preserving cryptographic algorithms, and the operational characteristics of Layer-2 zk-rollup architectures. The subsequent *System Design & Architecture* phase will formalise the ZKER design, defining all Layer-1 and Layer-2 components, their interactions, and the cryptographic primitives

required to support voter anonymity, ballot secrecy, and verifiable tallying.

During the *Smart Contract Prototype* phase, the core voting contracts, eligibility verification structures, and Layer-2 storage mechanisms will be implemented and deployed on the Polygon zkEVM testnet. The *ZK Circuit Integration* phase will extend this prototype by constructing and integrating zero-knowledge proof circuits for voter authentication, nullifier generation, ballot correctness, and homomorphic tally validation using frameworks such as Circom and SnarkJS. Completion of this end-to-end prototype is marked by the *Interim Report* milestone.

The *Evaluation & Testing* phase will empirically benchmark the ZKER system against a simulated Layer-1 baseline, measuring throughput, transaction cost reduction, latency improvements, proof verification behaviour, and overall scalability. Specific data preparation protocols will be implemented for each benchmarking dimension:

*Throughput Assessment Data.* Synthetic vote batches (1,000–20,000 transactions) will be generated to emulate realistic election loads, with each vote containing an encrypted ballot, a nullifier, and a ZK proof. These batch sizes remain computationally practical for Circom/SnarkJS and allow reliable measurement of Layer-2 batching capacity. Larger batches (50k–100k) are excluded due to slow proof generation, memory limits, and long processing times, while national-scale tests (100k–1M) are not feasible. Scalability at these sizes will instead be estimated through performance extrapolation and comparison with published zk-rollup benchmarks.

*Transaction Cost Evaluation Data.* Gas usage for vote submissions, contract writes, proof verification, and batch commitments will be measured on both Layer-1 and Layer-2, with multiple runs used to capture variance and derive average per-vote and per-batch costs. While this evaluation is fully feasible on testnets, it cannot replicate real Ethereum congestion, making congestion-driven gas modelling outside the project scope. Additionally, very large proof verification steps may approach L1 block gas limits, and testnet gas variability reduces the reliability of congestion-sensitive cost data.

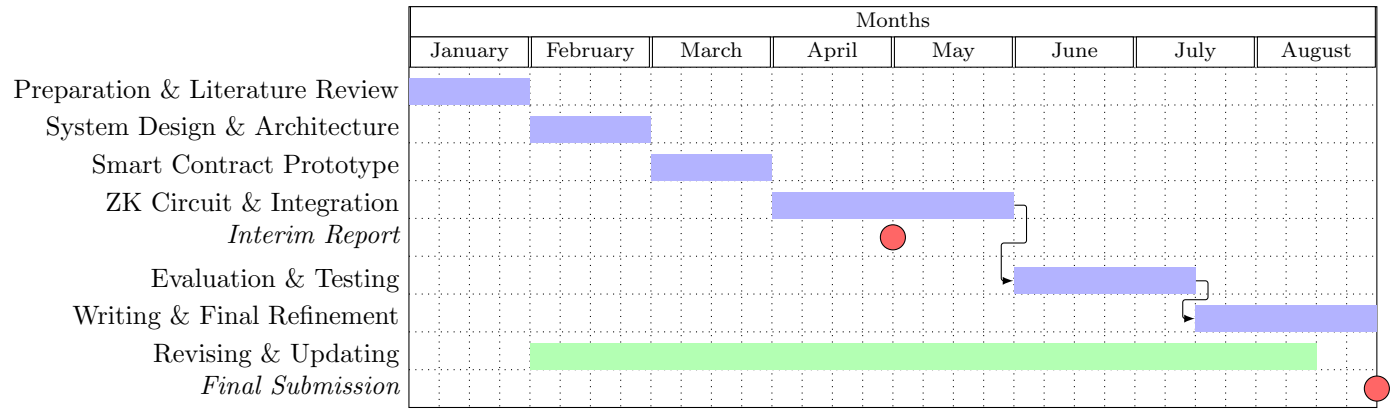
*Latency Measurement Data.* Timing data across the full transaction lifecycle and submission delay, batching, proof generation, and Layer-1 confirmation—will be collected to assess end-to-end responsiveness. Measurements are feasible for small and medium batches, but large batches (> 10,000 votes) are impractical due to slow proof generation and fluctuating Polygon zkEVM testnet confirmations. Election-scale latency cannot be evaluated, as testnets do not reflect real Layer-2 congestion, placing large-scale testing outside the project scope.

*Proof Verification Behaviour Data.* Valid and intentionally invalid proofs will be generated to assess verification robustness, including cases such as incorrect Merkle paths, duplicate nullifiers, and malformed ciphertexts. These tests are feasible at moderate scale, though generating large sets of invalid proofs requires custom tooling and some malformed inputs may expose verifier edge cases. Large-scale adversarial or fuzzing-based attacks cannot be evaluated, as they require specialised cryptographic analysis and infrastructure, placing them outside the project scope.

*Scalability Evaluation Data.* Stress-test datasets containing tens of thousands of encrypted ballots will be used to analyse how system performance scales with voter volume.

If time remains within the project schedule, for the *ZKP Scheme Comparison* (RQ4), an extended investigation may involve evaluating additional proof systems—such as zk-SNARKs, zk-STARKs, and Bulletproofs—either by constructing small prototype circuits or by analysing their proving and verification characteristics using established benchmark datasets. Such an extension would provide a more comprehensive understanding of how different proof systems influence proving time, verification overhead, and proof size in the context of ZK-Rollup-based voting. Similarly, for the *Layer-2 Environment Comparison* (RQ5), the research may be expanded to include limited experimental deployment on an alternative zk-Rollup platform, such as zkSync Era. Even a minimal implementation, for example deploying simplified smart contracts or conducting basic proof-verification tests, would offer practical comparative evidence to supplement the primarily literature-based architectural evaluation.

Finally, the *Writing & Final Refinement* phase will synthesise all results, complete the dissertation chapters, and refine the final document. In parallel, a continuous *Revising & Updating* track will ensure the work incorporates the latest developments in zk-rollup systems, zero-knowledge proof frameworks, and cryptographic voting protocols, maintaining technical relevance and academic rigour throughout the research process.



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